

1. Answer the following limit questions showing full justification.

a) $\lim_{x \rightarrow \infty} \frac{x^2 + x + 1}{x}$

b) $\lim_{x \rightarrow \infty} \frac{1 - 2x^2}{x^2 + 3x}$

2. Create three limits that all have the form of $\frac{\infty}{\infty}$ but produce the answers of $\frac{5}{2}$, 0, and ∞ .

3. Kyle is trying to answer the following questions and presents the following work to his professor. Why is Kyle's work incorrect or insufficient to justify his answer?

a) $\lim_{x \rightarrow \infty} \frac{x^2 + 1}{x + 1 + \frac{x^2}{2}} = \frac{\infty}{\infty} = \infty$.

b) $\lim_{x \rightarrow 3} \frac{x^3 + 3x + 3}{1 - x^2} = \lim_{x \rightarrow 3} \frac{x^3 + 3x + 3}{1 - x^2} \cdot \frac{1}{\frac{1}{x^2}} = \lim_{x \rightarrow 3} \frac{1 + \frac{3}{x} + \frac{3}{x^2}}{\frac{1}{x^2} - 1} = \lim_{x \rightarrow 3} \frac{1 + 0 + 0}{0 - 1} = \frac{1}{-1} = -1$.

4. Evaluate the limit $\lim_{x \rightarrow -\infty} \frac{\sqrt{x^2 + 1}}{x - 1}$ providing full justification.

5. Determine the horizontal and vertical asymptotes to $f(x) = \frac{3e^x + 4}{2e^x - 1}$

6. Determine the horizontal and vertical asymptotes to $f(x) = \begin{cases} \frac{2x}{x+1}, & x < -1 \\ \ln(x+1), & -1 < x \leq 5 \\ \cos(x), & x > 5 \end{cases}$.

Recall the limit laws that we discussed in class.

Theorem: Suppose that c is a constant, n is a positive integer and the limits $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exist.

Then,

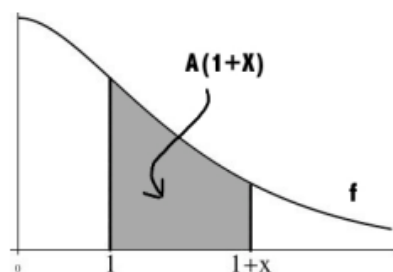
$$3. \lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$$

$$6. \lim_{x \rightarrow a} [f(x)g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$$

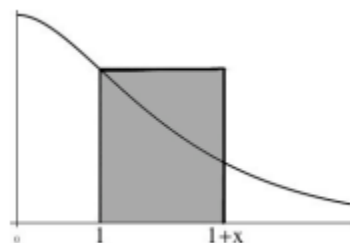
7. TRUE or FALSE: If neither $\lim_{x \rightarrow a} f(x)$ nor $\lim_{x \rightarrow a} g(x)$ exists, then $\lim_{x \rightarrow a} [f(x) + g(x)]$ must not exist. If the statement is true provide a justification demonstrating that it is true. If the statement is false, provide a counter-example that shows the statement is false.
8. TRUE or FALSE: If $\lim_{x \rightarrow a} [f(x) \cdot g(x)]$ does not exist then it must be the case that either $\lim_{x \rightarrow a} f(x)$ or $\lim_{x \rightarrow a} g(x)$ do not exist. If the statement is true provide a justification demonstrating that it is true. If the statement is false, provide a counter-example that shows the statement is false.

Now we turn to some more atypical questions to challenge our understanding of the topics of this chapter thus far.

9. The function f is continuous and decreasing on $(0, \infty)$. Define $A(1+x)$ to be the area from 1 to $1+x$ under the graph of f and above the x -axis (as shown in the picture below).



Compute $\lim_{x \rightarrow 0^+} \frac{A(1+x)}{x}$. Hint:



10. Is there a value of a such that $\lim_{x \rightarrow 3} \frac{2x^2 - 2ax + x - a - 1}{x^2 - 2x - 3}$ exists?